

NUCLEI

$$\begin{matrix} A \\ Z \end{matrix} X$$

$A \rightarrow$ Mass numbers
 $Z \rightarrow$ Atomic number
 (No. of protons)

Nucleone

Protons +
Neutrons

$$\text{No. of neutron} = A - Z$$

- $m_p = 1.66 \times 10^{-27} \text{ kg} = 1.0073 \text{ a.m.u}$
- $m_n = 1.67 \times 10^{-27} \text{ kg} = 1.0087 \text{ a.m.u}$
- $m_e = 9.1 \times 10^{-31} \text{ kg}$

$$m_\alpha > m_n \approx m_p > m_e$$

$$1 \text{ a.m.u} = 1.66 \times 10^{-27} \text{ kg} = \left(\frac{6 \text{ C}^{12}}{12} \right)$$

- * Isotopes - Same Z
- Isobars - Same A
- Isotones - Same $(A - Z)$

Rest Mass Energy

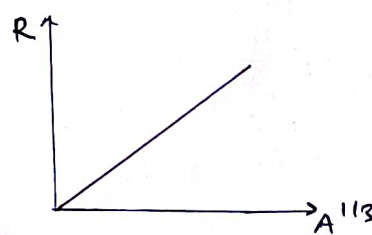
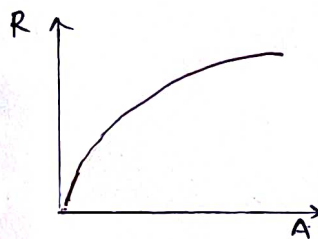
$$E = mc^2$$

$$E = 1 \text{ (a.m.u} \times c^2) = 931.5 \text{ MeV}$$

Nuclear Size

$$R = R_0 A^{1/3}$$

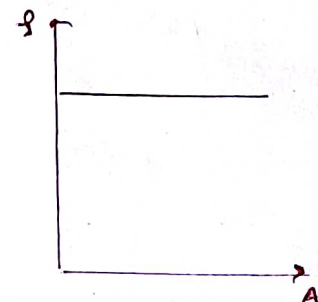
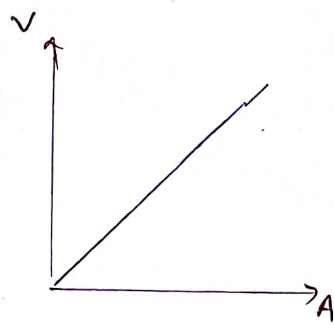
$$R_0 = 1.2 \times 10^{-15} \text{ m} = 1.2 \text{ fm}$$



Nuclear density

$$\rho = 10^{17} \text{ kg/m}^3$$

$\rightarrow$ constant
(Same for all nuclei)



Mass defect

$$\Delta m = Zm_p + (A - Z)m_n - M_{\text{nucleus}}$$

Binding energy of nucleus

$$\Delta E = (\Delta m)c^2 = (\Delta m) \times 931.5 \text{ MeV}$$

Released energy when nucleus is formed.

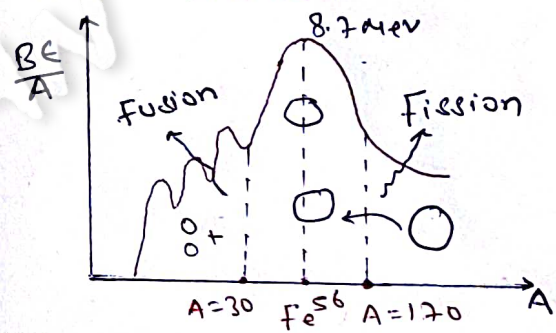
$$BE \propto \frac{B.E}{A} \text{ (Binding Energy per mass no.)}$$

Q $\rightarrow$ B.E of nuclei ${}^6\text{X}^{12}$ & ${}^4\text{Y}^6$ are 36 eV & 24 eV. Which are more stable?

$$\text{Sol}^n \left[\frac{B.E}{A} \right]_{\text{for X}} = \frac{36 \text{ eV}}{12} = 3 \text{ eV}$$

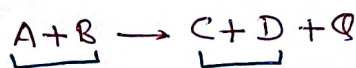
$$\left[\frac{B.E}{A} \right]_{\text{for Y}} = \frac{24 \text{ eV}}{6} = 4 \text{ eV} \text{ (more stable)}$$

* Average B.E of a nucleon inside a nucleus $\approx 8 \text{ MeV}$.



- As Mass number $\uparrow$ see, stability $\left(\frac{B.E}{A} \right)$ increases then decreases.
- Nucleons for lower masses $\rightarrow$ Fusion, (Energy released).
- Nucleons for Higher masses $\rightarrow$ Fission.
- Stability is high for $30 \leq A \leq 130$.
- $\left(\frac{B.E}{A} \right)_{\text{max}} = 8.7 \text{ MeV}$ for Fe
- $\left(\frac{B.E}{A} \right)_{\text{Avg}} = 8 \text{ MeV}$

Q-value of Nuclear Reaction



Reactant Product

Energy conservation

$$(m_A + m_B)c^2 = (m_C + m_D)c^2 + Q$$

$$Q = [(m_A + m_B) - (m_C + m_D)]c^2$$

$$Q = \left[\frac{\text{Mass of reactant}}{\text{Mass of product}} - \text{Mass of product} \right] \times c^2$$

$$Q = [(B.E.)_{\text{product}} - (B.E.)_{\text{reactant}}]$$

$$Q = [(B.E.)_C + (B.E.)_D - [(B.E.)_A + (B.E.)_B]]$$

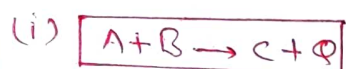
★ Value:-

★ $(m_A + m_B) > (m_C + m_D)$

$Q = +ve$, exothermic
Energy released.

★ $(m_A + m_B) < (m_C + m_D)$

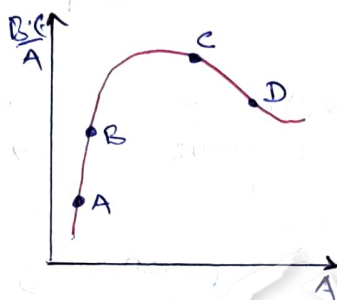
$Q = -ve$, endothermic
Energy absorbed.



$Q = +ve$, Energy released



$Q = -ve$, Energy absorbed

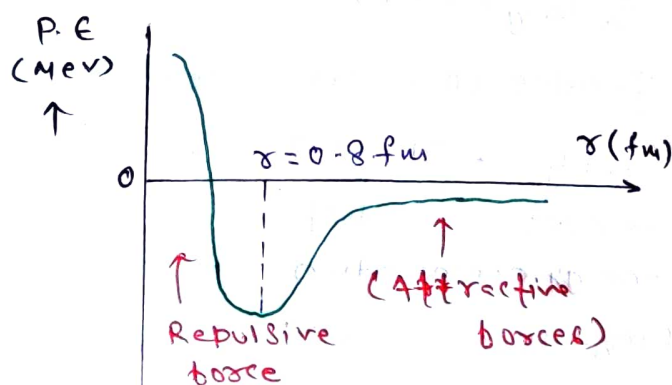


Nuclear force

- Strongest force in nature
- Short range, non-central, non-conservative, attractive force.
- They get saturated.
- charge dependent

$$F_{PP} = F_{PN} = F_{NN}$$

- Spin dependent
Like spin $\rightarrow$ less nuclear force
Unlike spin $\rightarrow$ more nuclear force
- does not follow inverse square law



Strong Nuclear force

- Strongest force
- Range $(10^{-15}m)$
- Always attractive
- Mediated meson

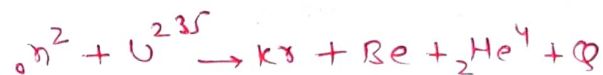
Weak Nuclear force

- Shortest Range $(10^{-16}m)$
- Repulsive
- Mediated by Bosons.

Nuclear fission

$\rightarrow$ Nuclear reactor in which a heavy nuclei breaks into two or more than two lighter nuclei.

$\rightarrow$ for nuclei stability



- Energy released per fission rxn in uranium is of the order

200 MeV per fissioning nucleus.

- ${}_{92}^{235}\text{U}$ & ${}_{94}^{239}\text{Pu}$ undergo fission with slow neutrons.

- Th^{232} undergo fission with fast neutrons.

Nuclear chain Reaction

$\rightarrow$ Avg. energy of a neutron produced in fission of ${}_{92}^{235}\text{U}$ is 2 MeV.

$$K = \frac{\text{Rate of prod}^n \text{ of neutron}}{\text{Rate of loss of neutron}}$$

$K = 1 \rightarrow$ critical

$K < 1 \rightarrow$ controlled

$K > 1 \rightarrow$ uncontrolled

Nuclear Reactor

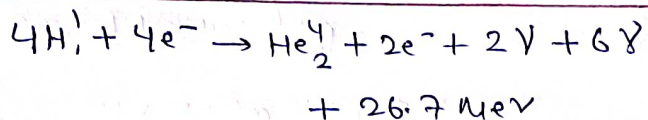
1. Fuel:- ${}_{92}^{235}\text{U}$ (fissionable uranium)
2. Moderator:- Graphite, water or D_2O .
3. Control Rods:- Cadmium (or) Boron rods.

Power of a Reactor

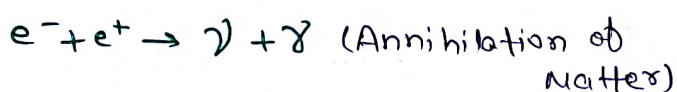
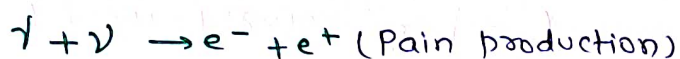
$$P = \frac{nE}{t} = \frac{nMc^2}{t}$$

Nuclear fusion

→ Process in which two or more than two lighter nuclei combine to form a heavier nucleus.



(Reaction inside sun)



Note: * Energy released per nucleon is more in fusion.

* Temp. of order 10^7 K are required for fusion.

Distance of closest approach when a particle is bombarded towards a heavy nucleus

$$r_0 \propto \frac{1}{m} \quad r_0 \propto \frac{1}{\sqrt{2}}$$